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PARAMETRIC-METRIC

Example

Scenario: Imagine we have a dataset of different countries. For each country, we have information on GDP, education level, access to clean water, and vaccination rates. We also have the average life expectancy for each country.

Goal: For a new country (or a region within a country) with known GDP, education level, access to clean water, and vaccination rates, we want to predict the life expectancy (Continuous or Discrete).

$$\text{Life Expectancy} = \beta_0 + \beta_1 \times \text{GDP} + \beta_2 \times \text{Clean Water Access} + \beta_3 \times \text{Education Level} + \beta_4 \times \text{Vaccination Rate} + \epsilon$$

- ▶ For example, β_3 is the coefficient for Education Level, indicating the change in life expectancy for a unit change in education level (however that is quantified, e.g., average years of schooling), while holding the other factors constant
- ▶ ϵ is the error term, capturing the variability in life expectancy not explained by the above variables.

Parametric Method

PARAMETRIC VS NON-PARAMETRIC MODELS

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